

• ASSEMBLYAI VOICE AGENT HACKATHON · 2026

VoiceChessmate

Chess you play entirely by voice.
No screen. No mouse. No sighted help.

THE PROBLEM

Chess is accessible. Chess **software** is not.

Chess is one of the few competitive arenas where blind and sighted players meet on genuinely equal terms. The IBCA has run world championships for decades. The game itself needs no sight at all.

43M

people are blind worldwide

295M

live with moderate-to-severe visual impairment

64

empty cells — what a screen reader announces
on a mainstream chess board

Sources: Lancet Global Health / GBD Vision Loss Expert Group prevalence estimates.

WHY TODAY'S OPTIONS FAIL

Every existing route is a **compromise**.

APPROACH	WHAT IT COSTS THE PLAYER
Mainstream platforms Chess.com, Lichess	Built around a drag-and-drop board. Screen readers read a grid, not a position. Fighting the interface, not the opponent.
Physical board + helper	Requires another person to be present. No remote opponents. No independence.
Braille / tactile board	Works beautifully in person. Cannot connect you to anyone online.
Text / notation interfaces	You must hold the entire position in memory, with no way to ask a question about it.
VoiceChessmate	Speak the move. Ask why it was bad and get the engine's real evaluation. Play blindfold, or hands-free while you walk. Things a board cannot do.

THE SOLUTION

Remove the interface entirely.

YOU SAY

"Knight to Felix 3"

THE COACH ANSWERS

"Knight f3. I'll play d5, challenging the centre."

AND WHEN YOU BLUNDER

"That was a blunder. Queen to Felix 7 gives up about 7.8 pawns. Nc3 was stronger."

Every answer is computed from the real position by a deterministic engine — never improvised by a language model.

ARCHITECTURE

The model talks. The **engine** decides.

Microphone → **AudioWorklet** — 25ms buffers, PCM16 at 24kHz

Dual AudioContext: capture at 24kHz to match the API, playback at the hardware's native 48kHz — forcing 24kHz output silently mutes the stream on PulseAudio/PipeWire.

▼ WebSocket

AssemblyAI Voice Agent API — Universal-3 Pro STT, turn detection, LLM routing, voice output

Keyterm biasing registers the IBCA phonetic vocabulary so the recogniser expects chess language.

▼ tool.call

18 typed JSON-schema tools — apply_move, describe_board, get_legal_moves, get_hint, undo_move, set_premove, resign_game ...

The agent may only act through these. It cannot assert a board state.

▼

Deterministic chess engine — chess.js rules + minimax, authoritative

Castling, en passant, promotion, repetition and the fifty-move rule are enforced here, not remembered by a model.

THE INSIGHT

Chess notation is hostile to speech recognition.

"b", "d", "e" and "g" are nearly indistinguishable over a microphone — and that is exactly the alphabet chess uses. One misheard letter is a legal, different, losing move.

"e4"

Four letters that sound alike. An ASR error here is invisible and unrecoverable.

"Eva four"

Phonetically distinct. The standard blind players already use in official tournaments.

Anna · Bella · Cesar · David

Eva · Felix · Gustav · Hector — the IBCA/FIDE file alphabet, spoken and understood in both directions

Not a gimmick: it is the notation the target user already knows, and it doubles as a phonetically separated vocabulary for the recogniser.

Prefer making **no move** over a wrong one.

A sighted player who sees a wrong move undoes it instantly.
A blind player may not discover the error for several moves
— by then the game is gone. So uncertainty refuses.

Confidence gates

A move plays untouched only at full confidence.
Anything less asks first. Ambiguity reads back
the real legal options.

Named piece is a constraint

If you say bishop, nothing but a bishop can move
— even when another piece could legally reach
the square.

Silence beats a guess

When an outcome can't be determined from the
position, the app says nothing rather than
announce a default.

507

tests across 22 files

~6k

lines of tests against ~10k lines of source

7

bugs logged with root cause — 5 carry
regression cases, open gaps documented rather
than hidden

MARKET

A market nobody is **serving**.

TAM

295M

people with moderate-to-severe visual impairment worldwide.

SAM

~4M

working-age, connected, in markets we can support at launch — applying a ~10% chess-participation assumption.

SOM · 3yr

~60K

reachable through blind associations, schools and chess federations, which are organised and easy to address.

SAM and SOM are bottom-up estimates from public prevalence data and a stated participation assumption, not measured demand. They are presented as a model to be validated, not a forecast.

Why now

Real-time voice agents only became low-latency and affordable enough for conversational turn-taking in the last ~18 months. This product was not buildable before.

BUSINESS MODEL

Four ways this pays for itself.

STREAM	WHO PAYS	WHY THEY PAY
Free tier	Nobody	Full play against the engine. Accessibility is never the paywall — that would defeat the product.
Pro subscription	Individual players	Tournament mode, opening trainer, spoken tactics drills, game analysis, unlimited coach conversation.
Platform licensing	Chess platforms	The accessibility gap is theirs to close and expensive to build. We are the missing voice layer, ready to embed.
Institutional	Schools for the blind, national associations, rehab centres	Site licences and training. These buyers have budgets, procurement paths, and a mandate.

Marginal cost per active player is streaming voice minutes, so cost control is a product feature, not an afterthought. Every session token is minted against a per-identity daily budget held in Postgres and a deployment-wide daily ceiling, both decremented under a row lock. Sessions reserve pessimistically and refund what they do not use. Spend is attributable to an identity through a ledger. When the ceiling is reached, voice pauses and the app stays fully playable by keyboard — a degradation, not an outage.

STATUS & ROADMAP

Shipped, tested, and **live**.

Today	Full game by voice or keyboard. Spoken move analysis, blindfold training, hands-free play, narrated famous games. Accounts, saved games, ratings. Live, 507 tests, CI-gated, MIT.
Next	Human-vs-human play over WebRTC · PGN import/export by voice · Spanish and German IBCA phonetic sets
Then	Opening trainer and tactics drills as spoken exercises · OTB tournament companion mode
Longer	License the voice layer to existing chess platforms

Chess never needed sight.
Its **software** just assumed it.